



Presentation Guide: Single-Pass Drone Video to Accurate 3D Model Generation System

Problem Statement ID: SIH26158 | Smart India Hackathon Prototype

1. Executive Summary (The 30-Second Elevator Pitch)

"Our project solves **SIH26158** by taking a **single, unconstrained drone flyover video** (not a pre-planned grid survey) and automatically converting it into an accurate, georeferenced 3D point cloud model.

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The system automates video keyframe extraction, masks out moving dynamic objects (vehicles/humans), recovers shadow details using CLAHE contrast enhancement, ingests embedded GPS telemetry, and executes GPU-accelerated **Structure-from-Motion (SfM)** and **Multi-View Stereo (MVS)** via COLMAP on an RTX 3050 GPU, outputting an interactive 3D WebGL Dashboard with Architectural White Clay Mode, 3D Semantic Segmentation, and click-to-measure height analytics."

2. Key Challenges & Our Technical Solutions

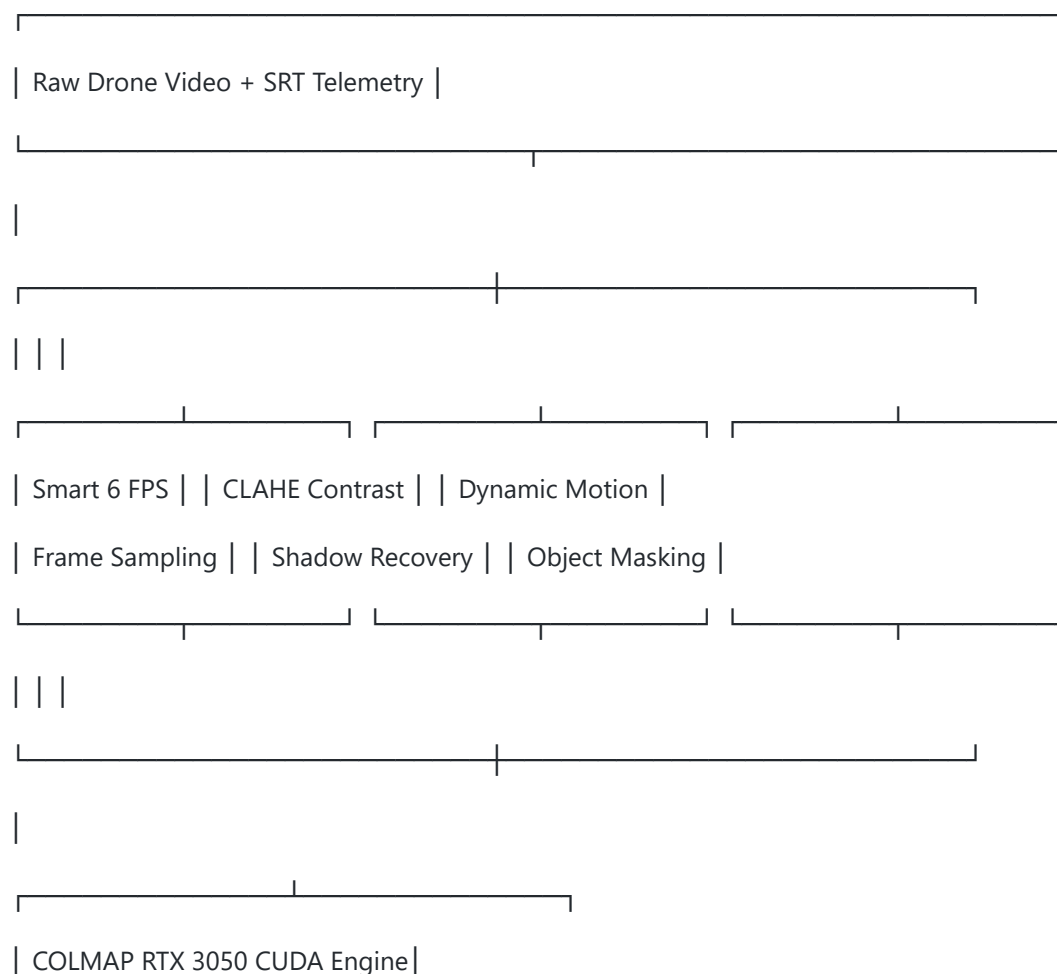
Key Problem Challenge (SIH26158)	Technical Solution Implemented
(i) Single Flight Path (Limited Viewing Angles)	High-Overlap Sampling (6 FPS): Decimates video to ~100 keyframes with ~90% spatial overlap.
(ii) Motion Blur & Compression Artifacts	Laplacian Variance Filtering: Automatically rejects blurry keyframes scoring below sharpness threshold.
(iii) Illumination & Shadows	CLAHE Contrast Recovery (<code>enhance_contrast.py</code>): Restores feature details in shadowed facades.
(iv) Dynamic Objects (Vehicles/Humans)	Motion Object Masking (<code>mask_dynamic_objects.py</code>): Generates SIFT feature masks to filter moving traffic.
(v & viii) GPS Inaccuracies & Metric Accuracy without GCPs	UTM Pose Priors & Georeferencing (<code>georeference_model.py</code>): Transforms WGS84 GPS to UTM cartesian pose priors.
(vi) Processing Requirements	RTX 3050 CUDA Acceleration: 16k SIFT features & 4K PatchMatch Stereo GPU execution.

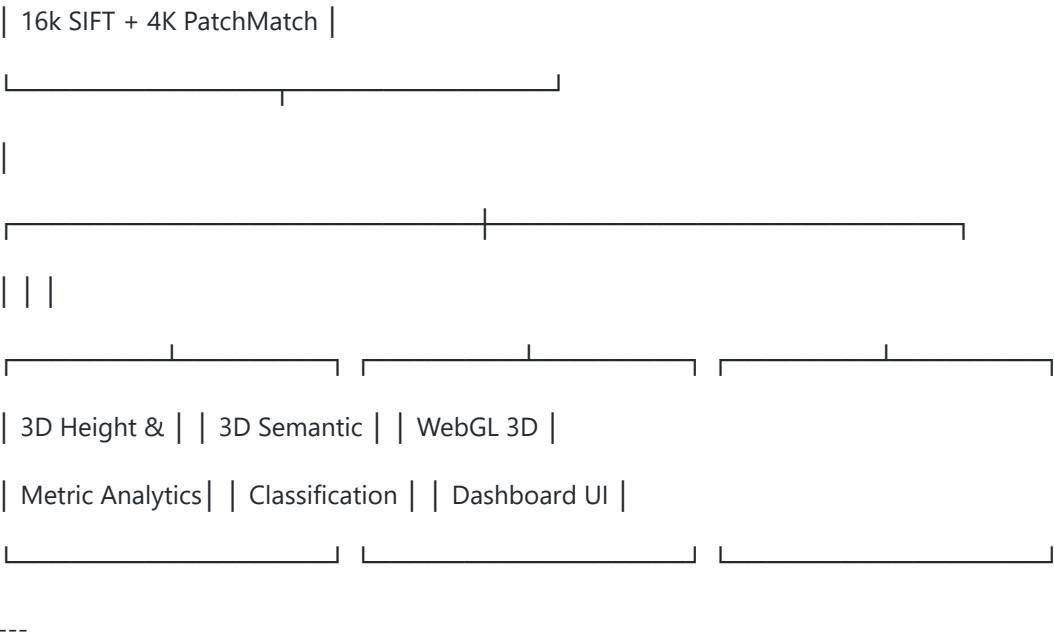
3. Desired Output & Evaluation Criteria Matrix

Deliverable / Evaluation Criterion	Expected Outcome	System Result Achieved
(i) 3D Terrain & Structures	Metrically accurate 3D model of building facades and roofs	558,595 dense 3D points & solid Octree Depth 12 Poisson mesh
(ii) Building Height Analytics	Calculate ground elevation & building height	25.79 meters structure height calculated automatically
(iii) 3D Semantic Classification	Classify buildings, roads, and vegetation	Multi-class 3D segmentation (Red: Buildings, Gray: Roads, Green: Trees)
(iv) Georeferencing	Real-world UTM metric coordinates	Georeferenced WGS84 UTM spatial placement
(v) Interactive Visualization	Multi-mode WebGL viewer with distance tools	White Clay Mode , Elevation Heatmap, RGB Color & Click-to-Measure Tool

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4. End-to-End System Architecture





5. Potential Applications

1. **Disaster Damage Assessment:** Near real-time 3D building height & collapse analysis.
2. **Military Reconnaissance & Border Mapping:** Single-pass flyover terrain mapping without pre-planned grid surveys.
3. **Urban Planning & Digital Twins:** High-accuracy 3D city models with automatic height extraction.
4. **Infrastructure Inspection:** Rapid inspection of bridges, highways, and high-rise structures.